UNIVERSITY OF RAJSHAHI
DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
B.Sc. Engineering Part III Even Semester Exam -2024
CSE 3242: Operating Systems Lab.
SET A

1. Write a multithreaded program that calculates various statistical values for a list of numbers. This program will be passed a series of numbers on the command line and will then create three separate worker threads. One thread will determine the average of the numbers, the second will determine the maximum value, and the third will determine the minimum value.

   Example: Suppose your program is passed the integers 90 81 78 95 79 72 85. The program will report:
   Output from Thread
   The average value is 82, Minimum value is 72, Maximum value is 95

2. Write a C program to implement **Shortest Job First (SJF)** (Non-Preemptive) scheduling. Your program should
   i. Accept the number of processes, Arrival Time, and Burst Time.
   ii. Sort processes by burst time (considering arrival times).
   iii. Calculate and display Waiting Time and Turnaround Time for each process, along with the averages.
   iv. Print the Gantt chart (order of execution).

3. Write a C program to implement the **FIFO** page replacement algorithm. Your program should:
   i. Accept the number of frames and the reference string (sequence of page numbers).
   ii. Calculate and print the total number of page faults.
   iii. Display the contents of the frames after each page reference.

CamScanner